

SkillBridge AI

"From Skill Gaps to Career Opportunities"

Live Web App: <https://prakalyapbl-create.github.io/skillbridge-ai/>

Target Audience: College Students, Fresh Graduates & Campus Placement Teams

Core Workflow Loop:

PROFILE ■■■ SKILL GAP ■■■ LEARN ■■■ PRACTICE ■■■ RESUME ■■■ MATCH ■■■ APPLY

■ Slide 2: Problem Statement

WARNING: Key Challenges Faced by 80%+ of College Students:

- **Skill Ambiguity:** Students do not know what specific technical skills they are missing for target roles (e.g. Software Engineer Intern).
- **Tutorial Hell & Resource Confusion:** Overwhelmed by thousands of unverified YouTube videos without structured paths.
- **Language Barriers:** High quality tech resources are predominantly in English; students preferring Tamil or Hindi get left behind.
- **Resume Keyword Stuffing:** Students struggle to describe projects authentically and fall prey to fake resume generators.
- **Opportunity Discovery Noise:** Difficulty finding genuine company internships vs scam portals.

■ Slide 3: The Solution — Continuous Career Readiness Loop

SkillBridge AI is **NOT** just a static job portal or standard chatbot. It creates a continuous 7-step career development cycle:

Step	Phase	Details & Functionality
1	STUDENT PROFILE	Collects degree, year, verified skills & resume upload without qualification fabrication.
2	SKILL GAP ANALYZER	Compares profile vs target role. Categorizes GREEN, YELLOW, RED skills.
3	LEARN NAVIGATOR	Recommends verified learning resources in Tamil, English, and Hindi with trust badges.
4	WEEKLY ROADMAP	Step-by-step weekly milestone targets syncing live with career readiness dashboard score.
5	PRACTICE PROJECTS	Practical student project recommendations with exact resume description guidance.
6	RESUME STUDIO	Mode A Builder + Mode B Resume Tutor AI & ethical job-specific tailoring.
7	MATCH & APPLY	Devpost hackathons, Unstop challenges & official company career portal matches.

Slide 4: Core Features Breakdown

1. Multilingual Support (Tamil, English, Hindi) • Seamless language switcher (EN, தமிழ், हिन्दी). • Preserves standard technical terms in English (Python, SQL, React, Git, REST API, DSA) while explaining concepts naturally in Tamil/Hindi.	2. Target Role & Skill Gap Analyzer Categorizes user skills against target roles: • GREEN (Strong): Verified matching proficiency. • YELLOW (Needs Improvement): Basic knowledge. • RED (Missing): Critical missing skill.
3. Personalized Learning Navigator with Provenance Badges Recommends verified courses with trust badges: • OFFICIAL: PostgreSQL Official Docs, MDN Docs. • GOVERNMENT / EDUCATIONAL: NPTEL / IIT Madras. • INDUSTRY OFFICIAL: GitHub Official Skills. • VERIFIED PLATFORM: FreeCodeCamp, CodeWithHarry.	4. Interactive Weekly Learning Roadmap Custom weekly milestone schedule with real-time status toggles (Not Started, Learning, Practicing, Completed ■), syncing dynamically with the career readiness dashboard score.

■ Slide 5: Practical Projects & Resume Studio

Practical Project Recommender

- Recommends student-friendly projects tailored to missing skills (e.g., SQL -> "Student Grade & Attendance Database").
- Includes objectives, technologies, expected learning, and exact resume bullet guidance.

Dual-Mode Resume Studio & Job Tailoring

- **Mode A ("Build my resume"):** Clean, formatted academic resume generator.
- **Mode B ("Resume Tutor AI"):** Interactive assistant teaching students how to write project descriptions and action verbs.
- **Job Tailoring:** Compares Job Description vs Resume with strict ethical safety guardrails ("Never invent experience. Only add if genuinely possessed").

■ Slide 6: Opportunity Discovery & CareerBot

Verified Opportunity Discovery & Matching

- Verified opportunities across Internships, Jobs, Devpost Hackathons, and Unstop Coding Challenges.
- Source tags: OFFICIAL COMPANY SOURCE vs THIRD-PARTY PLATFORM.
- Responsible matching score & requirement alignment breakdown.

CareerBot AI & Mind Refresh SkillBreak

- **CareerBot:** Contextual AI assistant connected to student profile details, target role, and preferred language.
- **Mind Refresh SkillBreak:** 3-minute educational mini-games (SQL bug hunter, Python code output predictor, professional communication exercises, HTTP trivia).

■ Slide 7: Technical Architecture

Technology Component	Purpose & Architectural Role
Next.js 14 (App Router)	Full-stack React framework with SSR and Static Export capability.
TypeScript	Strict type safety across all 16 interactive modules.
Tailwind CSS & Lucide	Modern, responsive student-friendly UI components.
Python FastAPI	Backend microservice for structured resume analysis.
GitHub Actions	Automated CI/CD deployment pipeline for GitHub Pages.

■ ■ Slide 8: Trust, Safety & Ethics Policy

IMPORTANT TRUST RULES:

- **Zero Qualification Fabrication:** SkillBridge AI strictly refrains from keyword stuffing or inventing fake experience.
- **Verified Provenance:** Every opportunity displays its verified source type and last verification date.
- **Responsible Phrasing:** Uses wording such as *"Your profile currently matches these listed requirements"* rather than fake selection guarantees.

■ Slide 9: Demo Case Study & Impact

Student Showcase: Arun Kumar

- **Degree:** B.E. Computer Science & Engineering (2nd Year, 2027)
- **Target Role:** Software Developer Intern
- **Preferred Language:** Tamil (ta)
- **Initial Skill Readiness:** 40% (Missing SQL, DSA, Git, REST APIs)

Result & Outcome:

Completed weekly SQL roadmap, built Student Attendance API project, updated resume bullet points, matched Zoho SDE Intern requirements with **85% readiness score**.

■ Slide 10: Conclusion & Presentation Links

- **Live Web App:** <https://prakalyapbl-create.github.io/skillbridge-ai/>
- **GitHub Repository:** <https://github.com/prakalyapbl-create/skillbridge-ai>

SkillBridge AI — Know What to Learn. Build What Matters. Be Ready for the Opportunity.